1 MWANGAZA ACADEMY – NEXUS SMIS ADVANCED ACADEMICS AND LEARNER REPORT CARDS TESTING GUIDE – 2027

Version: 1.0 · **Prepared for:** administrators, teachers, examination officers and QA testers · **School code:** MAK2027

> **Use this guide as a controlled test plan, not as an instruction to recreate or overwrite school data.** It covers only Academics—Assessments and Examination Management and Learner Report Cards.

1.1 Executive purpose

This continuation package tests whether the existing NEXUS SMIS installation can protect, calculate, review, approve, publish and reconcile Kenyan CBC assessment results and learner report cards. It deliberately separates configuration inspection from execution. A page that loads is not a pass: every result must be traced to an authoritative school-scoped record.

1.2 Environment preservation and live audit note

Item	Guide expectation	Read-only live audit / test instruction
School	MWANGAZA ACADEMY · MAK2027	Confirmed in the database: MWANGAZA ACADEMY, Kakamega, Lurambi, active.
Academic year	2027	Verify in Curriculum Setup before running any test; do not reset it.
Terms	Term 1: 6 Jan–2 Apr; Term 2: 27 Apr–6 Aug; Term 3: 30 Aug–30 Nov	Confirm dates and active-term badge in the live school; use the configured term as authoritative.
Learners	480 across PP1–Grade 9	Read-only audit found 110 current learners, 10 per grade, and only stream A records. Treat this as a configuration difference/defect to document, not something to silently repair in this test package.
Staff	24 teaching + 8 non-teaching	Read-only audit found 10 staff profiles for schoolId 60001.

Item	Guide expectation	Read-only live audit / test instruction
		Reconcile identities before assigning test responsibility.
Curriculum and allocations	Existing learning areas and teacher allocations	Read-only audit returned 0 school-scoped subjects and 0 active allocations. Do not import or create them from this workbook without approved remediation.
Assessment and reports	Use existing records	Read-only audit returned 0 assessments, 0 marks and 0 report cards for the audited school scope. Record the exact timestamp and query evidence.

Interpretation labels: `Working` means tested and reconciled; `Configured—untested` means visible but not exercised; `Partially functional` means some paths work; `Missing` means no authoritative record or control exists; `Defective` means behaviour contradicts the acceptance rule. The live audit above is evidence for triage only.

1.3 Operating rules before every test

1. Confirm the current-school badge reads MWANGAZA ACADEMY / MAK2027 and record a screenshot.
2. Confirm year, term and active-term badge; never infer a term from a browser URL or an old export.
3. Use a test learner or a clearly approved record. Do not overwrite existing production marks.
4. Download the latest NEXUS import template before any upload. The workbook delivered with this guide is a test dataset, not a claim that its headings are the current live importer contract.
5. Start with one learner, one learning area and one assessment; reconcile before expanding.
6. Capture actor, role, school, timestamp, selected filters, request result, record ID and screenshot for every pass or failure.
7. If a test needs new data, stop and obtain approval. Label the record `QA-2027-SUPPLEMENTAL` and never use it to disguise a live configuration gap.
8. Never directly manipulate the production database to simulate normal operations.

1.4 Section A — Pre-testing audit checklist

Audit item	Inspect	Where	Expected evidence	Status decision / response
Assessment categories and titles	Names, maximum marks, status, dates, year and term	Academics → Assessments; live query/read-only export	Screenshot + IDs + export	If absent, mark Missing; do not recreate without approval.
Academic year and terms	2027 and three date ranges; active term	Administration → Curriculum Setup	Year/term table + active badge	If dates overlap, raise High defect.
Grades, streams and enrolments	Actual names, streams, counts, learner IDs	People → Learners; Curriculum Setup	Roster export + counts	If guide differs, document difference and use live names.
CBC learning areas	Offered status, codes, max marks by grade	Curriculum Setup → Learning Areas	Subject list + grade mapping	Do not add unconfigured areas from the workbook.
Teacher allocations	Active teacher, class, subject, term and year	Allocation Register	Filtered register + teacher codes	Inactive/ replaced records must not authorize current entry.
Teacher accounts and roles	Admin, Deputy, Council, class and subject roles	Administration → Users	Role/access screenshots	Test server denial as well as hidden controls.
Grading scales	Percentage and CBC modes; sample boundaries	Curriculum Setup → Grading	Scale export + policy approval	Label boundaries illustrative unless approved by school.
Mark-entry records	Existing assessment/ mark IDs, status and last changer	Assessments → Marklists	Counts + sample IDs	Do not reset or bulk-replace existing rows.

Audit item	Inspect	Where	Expected evidence	Status decision / response
Report settings	Layout, logo, signatures, school identity and output options	Administration → Report Card Settings	Preview + saved settings	Missing identity fields are Medium/High depending on publication risk.
Comment banks	BE/AE/ME/ EE and class/ principal roles	Administration → Comment Settings	40-comment or configured-bank export	Class and Principal banks must remain distinct.
Approval workflow	Submit, return, approve, lock, publish and correction	Assessments / Reports	Status transition log	Any bypass of approval is Critical.
Permissions and privacy	School scope, role scope, report confidentiality	Users, API response and direct URL checks	Role matrix + denied response	Any cross-school data is Critical and blocks acceptance.

1.5 Section B — Academics: advanced test scenarios

1.6 40 executable Academics and Examination Management scenarios

The scenarios below are executable tests. For every case, capture the user role, school badge, selected year/ term, screen state, timestamp, screenshots, and resulting record identifier.

1.6.1 A01 — Create a custom assessment title

Steps. As Erick Bilishi, open Academics → Assessments, select 2027, Term 1, Grade 7 and one allocated learning area; enter "Opening Assessment"; save. **Expected result.** Only an authorized administrator can create it; the title, year, term, grade, subject, creator and timestamp persist. **Evidence.** Screenshot of form and saved record; audit event ID.

1.6.2 A02 — Create a second custom title

Steps. Create "Continuous Assessment Test 1" for the same period but a different configured assessment slot. **Expected result.** The system accepts a distinct title without changing the first record. **Evidence.** Two record IDs and an audit comparison.

1.6.3 A03 — Reject a duplicate assessment

Steps. Attempt to create the same title, period, grade and learning area twice. **Expected result.** A clear duplicate message appears and no second authoritative assessment is created. **Evidence.** Before/after record count and message.

1.6.4 A04 — Wrong academic year

Steps. Select 2026 while the test dataset is 2027 and try to create an assessment. **Expected result.** The wrong-year record is blocked or clearly marked as historical; no 2027 result is overwritten. **Evidence.** Selected filters and server response.

1.6.5 A05 — Wrong term isolation

Steps. Open a Term 1 assessment while Term 2 is active as a teacher. **Expected result.** The teacher is blocked; an authorized administrator sees the inactive-term confirmation before proceeding. **Evidence.** Warning text and audit event if overridden.

1.6.6 A06 — Maximum marks policy

Steps. Create or open an assessment with the approved maximum mark configured by the school. **Expected result.** The maximum is visible to the marker and is used consistently in percentage calculations. **Evidence.** Configuration screenshot and calculated example.

1.6.7 A07 — Unauthorized assessment creation

Steps. Sign in as Sarah Anupi or an ordinary subject teacher and look for Create New Assessment. **Expected result.** The control is hidden or disabled and the server rejects a direct request. **Evidence.** Role, UI state and rejected request.

1.6.8 A08 — Edit assessment metadata

Steps. As an administrator, change a title without changing its year, term, grade or subject. **Expected result.** The title changes while existing marks retain their assessment ID and history. **Evidence.** Before/after title and mark count.

1.6.9 A09 — Edit assessment with marks

Steps. Attempt to change a marked assessment's learning area or class. **Expected result.** The system requires a safe correction workflow or blocks the destructive change; marks remain traceable. **Evidence.** Prompt, audit entry, and unchanged mark IDs.

1.6.10 A10 — Deactivate an assessment

Steps. Deactivate a test assessment that has no production marks. **Expected result.** It becomes unavailable for new entry but remains visible in history. **Evidence.** Status before/after and list filter.

1.6.11 A11 — Allocated subject scope

Steps. As John Andera, open only learning areas and classes assigned to him. **Expected result.** Only active school-scoped allocations appear; another teacher's scope is rejected server-side. **Evidence.** Scope list and attempted unauthorized URL/API result.

1.6.12 A12 — Correct grade and stream

Steps. Use a teacher assigned to Grade 7 Blue or the configured equivalent. **Expected result.** The roster matches the selected grade/stream and does not include another stream. **Evidence.** Roster count and class label.

1.6.13 A13 — Save an ordinary mark

Steps. Enter 59 for Adrian Andayi in Kiswahili and save. **Expected result.** The mark persists and is available to the marklist and report-card read path. **Evidence.** Saved row, refresh result, and mark audit.

1.6.14 A14 — Save zero

Steps. Enter 0 for a learner who completed the assessment with zero. **Expected result.** Zero is treated as a valid score, not as missing. **Evidence.** Row status and exported value.

1.6.15 A15 — Save maximum

Steps. Enter 100 for a learner. **Expected result.** 100 is accepted, competency calculation is consistent, and no rounding changes it. **Evidence.** Marklist and report-card value.

1.6.16 A16 — Save decimal mark

Steps. Enter 59.5 where decimal marks are approved. **Expected result.** The system either preserves the supported precision or gives a clear validation message; it must not silently truncate. **Evidence.** Input, stored value, and message.

1.6.17 A17 — Reject above maximum

Steps. Enter 101 or a value above the configured maximum. **Expected result.** Inline/server validation rejects it and no invalid mark is stored. **Evidence.** Validation message and unchanged database count.

1.6.18 A18 — Leave a mark blank

Steps. Clear one learner's mark and save. **Expected result.** Blank remains missing, never zero; missing count increases. **Evidence.** Marklist missing badge and export.

1.6.19 A19 — Duplicate save

Steps. Submit the same learner row twice from two browser tabs. **Expected result.** The authoritative row is updated safely; duplicate records are not created. **Evidence.** Two request timestamps and one final mark row.

1.6.20 A20 — Missing learner detection

Steps. Compare the roster to saved marks after omitting one learner. **Expected result.** The whole-class marklist identifies the learner as missing. **Evidence.** Roster-to-mark reconciliation.

1.6.21 A21 — Stream transfer

Steps. Move a clearly identified test learner between configured streams using the approved workflow. **Expected result.** Historical marks remain intact and new entry follows the new class without cross-stream leakage. **Evidence.** Transfer history and before/after reports.

1.6.22 A22 — Post-submission edit

Steps. Submit a subject assessment, then try to edit as the subject teacher. **Expected result.** Teacher editing is blocked after submission; authorized correction follows the review workflow. **Evidence.** Status transition and rejected edit.

1.6.23 A23 — Concurrent teacher edits

Steps. Open the same assessment as two authorized users and save different values. **Expected result.** The final state is deterministic and the audit trail identifies both attempts/changes. **Evidence.** Audit records and final reconciliation.

1.6.24 A24 — Refresh persistence

Steps. Save marks, refresh the browser, sign out and sign back in. **Expected result.** Saved marks remain identical and are loaded from the database, not local UI state. **Evidence.** Before/after screenshots and query result.

1.6.25 A25 — Whole-class access — Super Administrator

Steps. Open Whole Class Marklist as Erick Bilishi. **Expected result.** The complete configured class period appears with all active learning areas, learners, missing indicators and exports. **Evidence.** Role badge, full table, export files.

1.6.26 A26 — Whole-class access — Administrator

Steps. Open the same view as the configured Administrator. **Expected result.** Administrator access is equivalent for authorized school records and remains school-scoped. **Evidence.** Role badge and cross-school negative test.

1.6.27 A27 — Whole-class access — Examination Council

Steps. Open the view as Daniel Barasa. **Expected result.** Examination Council can review authorized whole-class data without gaining unrelated configuration privileges. **Evidence.** Review screen and denied settings action.

1.6.28 A28 — Whole-class filter by term

Steps. Select Term 1, Term 2 and Term 3 in separate checks. **Expected result.** Only the selected term's records appear; no mixed-term columns or scores are shown. **Evidence.** Three exports with term labels.

1.6.29 A29 — Whole-class filter by stream

Steps. Filter to one configured stream. **Expected result.** Only that stream's learners and learning areas appear. **Evidence.** Filtered count and sample admission numbers.

1.6.30 A30 — Whole-class export and print

Steps. Sort by learner name, admission number and average; export/print after each sort. **Expected result.** Output preserves the visible order and includes school identity, term, class, learning areas and missing indicators. **Evidence.** PDF/print preview and workbook row order.

1.6.31 A31 — Reconcile subject and whole-class marks

Steps. Compare one subject marklist with the corresponding whole-class cells. **Expected result.** Every entered value matches by learner ID/admission number and term. **Evidence.** Reconciliation sheet and discrepancy count.

1.6.32 A32 — Examination Council unusual score review

Steps. Use a test set containing 0, 1, 59, 79.5 and 100. **Expected result.** Review workflow highlights unusual or missing values without changing them automatically. **Evidence.** Review notes and unchanged mark rows.

1.6.33 A33 — Moderation correction

Steps. Council flags a wrong submitted mark; administrator confirms a correction. **Expected result.** Corrected value is stored, original value remains traceable, and teacher is notified where configured. **Evidence.** Correction reason, audit chain, notification.

1.6.34 A34 — Approved result protection

Steps. Approve or lock an assessment and attempt a normal teacher edit. **Expected result.** Edit is blocked; only the designated correction function can change it. **Evidence.** Status, error message, audit record.

1.6.35 A35 — Percentage grading boundaries

Steps. Test illustrative boundary values 0, 10, 20, 21, 30, 31, 40, 41, 57, 58, 74, 75, 89 and 90. **Expected result.** Each maps to the configured sample scale consistently; the guide does not claim official KICD thresholds. **Evidence.** Boundary matrix export.

1.6.36 A36 — CBC competency display

Steps. Verify BE1, BE2, AE1, AE2, ME1, ME2, EE1 and EE2 across marklist and reports. **Expected result.** Labels and descriptions are consistent and missing remains blank. **Evidence.** Marklist/report screenshots.

1.6.37 A37 — Equal scores

Steps. Give two learners the same score. **Expected result.** They receive the same percentage and competency; sorting uses a stable tie-breaker. **Evidence.** Sorted rows and calculations.

1.6.38 A38 — Assessment with incomplete records

Steps. Generate analysis while some marks are blank. **Expected result.** Missing values are excluded from averages according to policy and clearly disclosed. **Evidence.** Analysis summary and missing count.

1.6.39 A39 — Learning-area exclusion

Steps. Use a learning area not offered to the selected grade. **Expected result.** It cannot be entered or silently added to the class marklist. **Evidence.** Configuration evidence and rejected attempt.

1.6.40 A40 — Historical retrieval

Steps. Open an earlier completed assessment after creating a newer one. **Expected result.** Historical records remain readable and are not replaced by the newest assessment. **Evidence.** Assessment IDs and report snapshots.

1.6.41 A41 — Audit completeness

Steps. Review mark create, save, submit, return, approve, correct and override events. **Expected result.** Each sensitive transition identifies actor, school, entity, timestamp and reason where applicable. **Evidence.** Audit-log extract and coverage checklist.

1.7 40 executable Learner Report Card scenarios

The scenarios below are executable tests. For every case, capture the user role, school badge, selected year/term, screen state, timestamp, screenshots, and resulting record identifier.

1.7.1 R01 — Select one assessment

Steps. As Sarah Anupi, select one approved assessment for a configured class and generate one learner report. **Expected result.** Only the selected assessment appears and its marks match the approved source. **Evidence.** Generated PDF and source marklist.

1.7.2 R02 — Select multiple assessments

Steps. Select Mid-Term and End-Term for one learner. **Expected result.** The report distinguishes assessment periods and does not blend terms. **Evidence.** Report headings and reconciliation.

1.7.3 R03 — Select all relevant assessments

Steps. Select all approved assessments for a term. **Expected result.** Only same-year, same-term assessments appear; duplicates are prevented. **Evidence.** Selection list and report.

1.7.4 R04 — Change selected assessments

Steps. Remove one assessment and regenerate. **Expected result.** Removed results disappear without deleting the source assessment. **Evidence.** Before/after PDFs.

1.7.5 R05 — Wrong-year selection

Steps. Attempt to combine 2026 and 2027 assessments. **Expected result.** The report generator blocks the mix or clearly separates historical context; no silent merge occurs. **Evidence.** Validation message.

1.7.6 R06 — Wrong-term selection

Steps. Attempt to combine Term 1 and Term 2 in one term report. **Expected result.** The term filter prevents the mix. **Evidence.** Selected filters and rejected generation.

1.7.7 R07 — Missing selection

Steps. Generate without selecting an assessment. **Expected result.** A clear actionable message appears and no misleading blank report is published. **Evidence.** Message and no-output evidence.

1.7.8 R08 — Marks flow through

Steps. Generate Adrian Andayi's report after saving Kiswahili 59, RE 44 and Creative Arts 44 in the test environment. **Expected result.** All relevant approved subjects appear with the exact values and competencies. **Evidence.** Source-to-report reconciliation.

1.7.9 R09 — BE1/BE2 display

Steps. Use illustrative scores in the BE range. **Expected result.** BE1 and BE2 display with correct descriptions, without claiming official thresholds. **Evidence.** Report page and grading sheet.

1.7.10 R10 — AE1/AE2 display

Steps. Use illustrative scores in the AE range. **Expected result.** AE1 and AE2 are distinct and consistently described. **Evidence.** Report page.

1.7.11 R11 — ME1/ME2 display

Steps. Use illustrative scores in the ME range. **Expected result.** ME1 and ME2 are distinct and consistently described. **Evidence.** Report page.

1.7.12 R12 — EE1/EE2 display

Steps. Use illustrative scores in the EE range. **Expected result.** EE1 and EE2 are distinct and consistently described. **Evidence.** Report page.

1.7.13 R13 — No mixed grading modes

Steps. Generate a percentage-based report and a CBC report using their intended settings. **Expected result.** The report does not show a percentage as a competency label or vice versa. **Evidence.** Two reports and settings.

1.7.14 R14 — Class-teacher comment selection

Steps. As a class teacher, select a predefined academic comment by category. **Expected result.** The comment is selected from the bank; arbitrary free typing is not required. **Evidence.** Selected comment and comment ID.

1.7.15 R15 — Class-teacher comment independence

Steps. Select a comment in Grade 4 and inspect the principal-comment field. **Expected result.** Principal comments remain unchanged and independently selectable. **Evidence.** Two field values.

1.7.16 R16 — Principal comment selection

Steps. As Peter Wekesa, select a conduct/development comment. **Expected result.** Principal comment bank is separate from academic comments and is recorded distinctly. **Evidence.** Role, comment category and saved report.

1.7.17 R17 — Comment category validation

Steps. Try to assign an EE comment to a BE learner without confirmation. **Expected result.** The system warns or requires an authorized choice; it must not silently mislabel the learner. **Evidence.** Validation message.

1.7.18 R18 — Missing comment handling

Steps. Generate a report with no class-teacher or principal comment. **Expected result.** The report identifies the missing section and follows the publication policy. **Evidence.** Draft report and warning.

1.7.19 R19 — Individual report

Steps. Generate one learner report by admission number. **Expected result.** The learner identity, grade, stream, year and term are correct. **Evidence.** PDF identity checklist.

1.7.20 R20 — Selected learners

Steps. Select a small group of learners and generate in one batch. **Expected result.** Only selected learners appear, each on the correct report. **Evidence.** Batch count and PDFs.

1.7.21 R21 — Whole-class reports

Steps. Generate the complete configured class. **Expected result.** Every learner appears once and no learner from another class is included. **Evidence.** Roster-to-PDF count.

1.7.22 R22 — Whole-grade reports

Steps. Generate a whole-grade report across the configured streams. **Expected result.** The output groups or labels streams correctly and preserves school scope. **Evidence.** Stream reconciliation.

1.7.23 R23 — Bulk PDF generation

Steps. Generate a single PDF for a selected class. **Expected result.** All pages render, page breaks are clean, and each report has correct identity. **Evidence.** PDF page count and sampled pages.

1.7.24 R24 — Bulk printing

Steps. Use print preview for a multi-learner batch. **Expected result.** The print layout is readable on A4 and does not truncate learning areas or comments. **Evidence.** Print preview screenshot.

1.7.25 R25 — Regenerate after correction

Steps. Correct one approved mark through the authorized workflow and regenerate. **Expected result.** Only the corrected value and dependent competency change; unrelated values remain unchanged. **Evidence.** Before/after report diff.

1.7.26 R26 — Incomplete marks

Steps. Generate a report with one missing subject mark. **Expected result.** The report states incomplete/missing status and does not convert blank to zero. **Evidence.** Report and source mark.

1.7.27 R27 — Transferred learner

Steps. Generate a report for a learner transferred between streams. **Expected result.** Historical report uses the correct historical class context and current records remain isolated. **Evidence.** Transfer history and report.

1.7.28 R28 — Historical report retrieval

Steps. Open a prior term report after a new term is active. **Expected result.** The prior term remains available and is not overwritten. **Evidence.** Term-labelled report.

1.7.29 R29 — School identity

Steps. Inspect report header/footer. **Expected result.** MWANGAZA ACADEMY, code MAK2027, approved school details and logo/settings appear where configured. **Evidence.** Header screenshot.

1.7.30 R30 — Learner identity fields

Steps. Inspect name, admission number, grade and stream. **Expected result.** All identity fields match the authoritative learner record. **Evidence.** Source row and report.

1.7.31 R31 — Approval information

Steps. Inspect draft, reviewed, approved and published states. **Expected result.** The report status and approver identity are visible according to permission policy. **Evidence.** Workflow history.

1.7.32 R32 — Teacher review

Steps. As a class teacher, review a draft report. **Expected result.** Review is allowed only for assigned class/role and does not grant unrelated admin rights. **Evidence.** Role test and access result.

1.7.33 R33 — Principal review

Steps. As Peter Wekesa, review an authorized report. **Expected result.** Principal can select principal comments and approve only where policy permits. **Evidence.** Role test and audit event.

1.7.34 R34 — Unauthorized publication

Steps. Attempt publication as an ordinary subject teacher. **Expected result.** Server rejects the action even if the UI is manipulated. **Evidence.** Rejected request and no status change.

1.7.35 R35 — Correction after approval

Steps. Attempt an ordinary edit after publication. **Expected result.** The system blocks it and provides a controlled correction path. **Evidence.** Message and unchanged report.

1.7.36 R36 — Confidentiality

Steps. Request a different learner's report by changing an ID in the URL/request. **Expected result.** School and role checks prevent unauthorized access and no data is returned. **Evidence.** Rejected response.

1.7.37 R37 — No cross-learner leakage

Steps. Generate two adjacent reports and compare all identity/mark sections. **Expected result.** Each page contains only its own learner's data. **Evidence.** Page-by-page checklist.

1.7.38 R38 — Export consistency

Steps. Compare on-screen, PDF and print outputs. **Expected result.** Marks, competency, comments, term and identity are identical across outputs. **Evidence.** Three-way reconciliation.

1.7.39 R39 — Performance summary

Steps. Generate class/whole-school subject-level summaries. **Expected result.** Totals, averages, missing counts and competency distributions reconcile to approved marks. **Evidence.** Summary export and formulas.

1.7.40 R40 — Refresh/sign-in persistence

Steps. Save a report comment, refresh, sign out and return. **Expected result.** The saved comment and status persist from database records. **Evidence.** Before/after evidence.

1.8 Section D — Integrated 20-day practical testing timetable

Day	Lead role	Assignment	Expected output / evidence
1	Administrator + QA	Freeze read-only configuration snapshot and school badge evidence.	Audit checklist, counts, screenshots, no mutations.
2	Administrator	Confirm academic year, term dates and active-term workflow.	Term table, active-term screenshot, defect entries.
3	Examination Officer	Reconcile grades, streams, learner rosters and admission numbers.	Roster counts and mismatch list.
4	Curriculum lead	Reconcile offered learning areas and grading settings.	Learning-area matrix; no unauthorized additions.
5	Administrator	Reconcile staff roles, teacher codes and allocations.	Allocation register and permission matrix.
6	Subject teacher	Run A11–A20 for one approved test learner and one area.	Saved marks, validations, refresh evidence.
7	Subject teacher	Run A21–A24: transfer, submission, persistence and concurrency.	Workflow screenshots and audit IDs.
8	Administrator	Run A01–A10 assessment lifecycle	Assessment records and duplicate checks.

Day	Lead role	Assignment	Expected output / evidence
		tests in a QA-approved scope.	
9	Examination Council	Run A25–A33 whole-class review, filters, export and moderation.	Whole-class export and reconciliation.
10	QA + Developer	Run A34–A40 grading, incomplete, historical and audit tests.	Boundary matrix and audit coverage.
11	Class Teacher	Run R01–R10 selection, term isolation and mark flow.	One learner report and source trace.
12	Class Teacher	Run R11–R18 CBC and comment-bank tests.	Comment selections and competency examples.
13	Principal	Run R15–R18 and R31–R35 principal review/security tests.	Principal comment and approval evidence.
14	Administrator	Run R19–R28 individual, selected, class, grade and bulk outputs.	PDF/print batch with page count.
15	QA	Run R29–R40 identity, privacy, regeneration and reconciliation.	Three-way output reconciliation.
16	QA + all roles	Run negative tests N01–N12.	Failure log and defect IDs.
17	QA + all roles	Run negative tests N13–N24.	Failure log and role-denial evidence.
18	QA + Developer	Run negative tests N25–N35; classify severity.	Defect register with owners.

Day	Lead role	Assignment	Expected output / evidence
19	Administrator + QA	Retest corrected High/Critical defects; verify no data loss.	Retest results, updated defect register.
20	Acceptance panel	Review acceptance checklist and sign only if all blocking defects are closed.	Signed acceptance record and evidence index.

1.9 Section E — Deliberate failure tests

Run the following cases only against approved QA records. The expected response must be specific, user-readable and backed by a server-side denial.

ID	Failure case	Test action	Expected behaviour
N01	Teacher creates assessment	Subject teacher clicks or calls create endpoint	Creation hidden/ rejected; no record created.
N02	Wrong school assessment access	Change school or entity identifier	Forbidden/not found; no data leakage.
N03	Above maximum mark	Enter 101	Inline/server validation; no save.
N04	Negative mark	Enter -1	Validation; no save.
N05	Malformed score	Enter text or NaN	Clear invalid-score message.
N06	Unauthorized subject	Teacher chooses unallocated subject	Scope rejected.
N07	Unauthorized class	Teacher chooses another stream	Scope rejected.
N08	Inactive-term teacher save	Teacher saves Term 1 while Term 2 active	Blocked with active-term warning.
N09	Inactive-term admin without confirmation	Admin saves past-term marks without confirmation flag	Confirmation-required error.
N10	Inactive-term override by Council	Examination Council attempts override	Rejected unless explicitly authorized.

ID	Failure case	Test action	Expected behaviour
N11	Duplicate assessment	Same period/class/ subject/title	Duplicate prevention.
N12	Duplicate learner rows	Upload same learner twice	Upload rejected or duplicate row clearly reported.
N13	Unknown admission number	Typed upload has unknown learner	Row-level error; no partial silent match.
N14	Wrong assessment type	Upload End-Term file to Mid-Term	File validation or explicit mapping required.
N15	Wrong term upload	Term 1 file selected for Term 3	Blocked with term mismatch.
N16	Blank required key	Admission and assessment number blank	Row-level required-key message.
N17	Approved mark edit	Teacher edits approved record	Blocked; correction workflow required.
N18	Locked mark edit	Normal save against locked record	Blocked and audited.
N19	Missing mark as zero	Blank cell uploaded	Remains missing, never zero.
N20	Wrong learner report ID	Alter report learner ID	Forbidden/not found.
N21	Cross-school report ID	Use a report card from another school	Forbidden/not found.
N22	Mixed-year report	Select 2026 and 2027	Blocked or explicitly historical-separated.
N23	Mixed-term report	Select Term 1 and Term 2	Blocked for single-term report.
N24	Publish without approval	Teacher publishes draft	Rejected.
N25	Principal comment as teacher	Teacher edits principal field	Rejected or field hidden.
N26	Wrong comment category	BE learner receives EE-only selection	Warning/validation.

ID	Failure case	Test action	Expected behaviour
N27	Report with no assessments	Generate empty selection	Actionable warning; no misleading output.
N28	Duplicate learning area	Two entries for same subject	Prevent duplicate column or flag defect.
N29	Concurrent conflicting save	Two users save different values	Deterministic result and audit trail.
N30	Delete marked assessment	Delete assessment with marks	Blocked or archival only; no data loss.
N31	Tamper export order	Change sort after export begins	Output either preserves captured order or reports state.
N32	Unauthenticated endpoint	Call assessment/ report API without session	Authentication required.
N33	Cross-role admin matrix	Ordinary admin invokes restricted super-admin action	Permission denied.
N34	Invalid competency code	Inject unsupported code	Validation rejects and does not persist.
N35	Comment injection/ overlong text	Paste script or overlong content	Sanitized/length-limited; no script execution.

1.10 Section F — Defect register template

Field	Required entry
Defect ID	Complete during execution
Module	Complete during execution
Test ID	Complete during execution
Description	Complete during execution
Severity: Critical / High / Medium / Low	Complete during execution
Steps to reproduce	Complete during execution
Expected result	Complete during execution
Actual result	Complete during execution
Evidence links / filenames	Complete during execution

Field	Required entry
Assigned developer	Complete during execution
Status	Complete during execution
Retest result	Complete during execution

Blocking rule: Critical security, data-loss, cross-school leakage, wrong learner identity, wrong term mixing or examination-result corruption blocks acceptance until fixed and retested.

1.11 Section G — Final acceptance checklist

Acceptance control	Pass condition	Evidence
Marks accuracy	Every sampled mark reconciles to the authoritative assessment row.	Sample reconciliation sheet.
CBC grading	All eight sample subcategories render consistently; approved school scale is used.	Boundary matrix + report samples.
Report calculations	Percentages, averages, missing values and competencies are reproducible.	Formula/output comparison.
Assessment selection	Only selected, same-year and same-term assessments are included.	Filter screenshots + reports.
Comments	Class Teacher and Principal banks remain separate and selected values persist.	Comment IDs + reports.
Permissions	Create, edit, approve, publish and override controls match role policy server-side.	Role test matrix.
Privacy	No learner or school data crosses scope; direct ID tampering is denied.	Denied requests.
History	Corrections and inactive-term overrides identify actor, time, school, entity and reason.	Audit export.
Bulk outputs	Class/grade batches contain correct learners once, readable on A4 and downloadable.	PDF and print evidence.

Acceptance control	Pass condition	Evidence
Reliability	Refresh, sign-out/sign-in and concurrent use preserve correct data.	Persistence/concurrency evidence.

1.12 Appendix A — 40 predefined Class Teacher comments

Category	Subcategory	Comment
BE	BE1	Is beginning to develop understanding and benefits from guided practice.
BE	BE2	Needs regular support to connect classroom ideas with practical tasks.
BE	BE1	Participates with encouragement and is developing confidence in communication.
BE	BE2	Can identify simple patterns when supported with examples and prompts.
BE	BE1	Is working towards greater independence in completing learning activities.
BE	BE2	Shows willingness to learn and should continue practising core skills consistently.
BE	BE1	Benefits from step-by-step instruction and frequent opportunities to explain thinking.
BE	BE2	Is developing collaboration skills and responds positively to constructive feedback.
BE	BE1	Needs continued reinforcement to apply knowledge beyond familiar examples.

Category	Subcategory	Comment
BE	BE2	Has made early progress; targeted support will help consolidate learning outcomes.
AE	AE1	Is approaching the expected outcome and can apply learning in familiar situations.
AE	AE2	Shows improving understanding and is beginning to explain solutions with clarity.
AE	AE1	Participates well and should continue building consistency across learning tasks.
AE	AE2	Uses practical skills with support and is developing greater independence.
AE	AE1	Can solve routine problems and is ready for more challenging applications.
AE	AE2	Communicates ideas increasingly clearly and benefits from constructive discussion.
AE	AE1	Demonstrates creativity in activities and should take more initiative in refining work.
AE	AE2	Works cooperatively and is learning to use feedback to improve performance.
AE	AE1	Is making steady improvement in knowledge, application and confidence.
AE	AE2	Would benefit from regular revision to strengthen transfer of learning.

Category	Subcategory	Comment
ME	ME1	Meets the expected outcome and applies knowledge accurately in most tasks.
ME	ME2	Demonstrates sound understanding, practical competence and responsible participation.
ME	ME1	Communicates ideas clearly and contributes positively to collaborative activities.
ME	ME2	Solves routine problems independently and explains methods appropriately.
ME	ME1	Produces consistent work and responds constructively to feedback.
ME	ME2	Applies learning to practical contexts with confidence and care.
ME	ME1	Shows good creativity and is willing to explore alternative solutions.
ME	ME2	Works independently, manages tasks well and meets agreed expectations.
ME	ME1	Participates actively and demonstrates dependable progress across the term.
ME	ME2	Has achieved the expected learning outcomes and should maintain this consistency.
EE	EE1	Exceeds the expected outcome and applies learning confidently in new situations.
EE	EE2	Demonstrates excellent understanding, practical

Category	Subcategory	Comment
		skill and purposeful communication.
EE	EE1	Solves complex problems independently and explains reasoning with precision.
EE	EE2	Consistently produces high-quality work and uses feedback to extend learning.
EE	EE1	Shows exceptional creativity and develops thoughtful, original solutions.
EE	EE2	Leads collaborative tasks responsibly while valuing the ideas of others.
EE	EE1	Applies knowledge with initiative, accuracy and strong attention to detail.
EE	EE2	Communicates clearly, listens respectfully and contributes maturely to discussions.
EE	EE1	Demonstrates sustained excellence, independence and commitment to improvement.
EE	EE2	Has exceeded the expected learning outcomes and is ready for enriched challenges.

1.13 Appendix B — 10 predefined Principal comments

ID	Role	Comment
P01	Head of Institution / Principal	Conducts self with respect and demonstrates positive personal development.
P02	Head of Institution / Principal	Shows responsibility and should continue strengthening punctuality and preparedness.

ID	Role	Comment
P03	Head of Institution / Principal	Participates positively in school life and cooperates well with others.
P04	Head of Institution / Principal	Demonstrates growing leadership and uses influence responsibly.
P05	Head of Institution / Principal	Is respectful, dependable and contributes to an orderly learning environment.
P06	Head of Institution / Principal	Shows good discipline and responds positively to guidance.
P07	Head of Institution / Principal	Works well with peers and demonstrates consideration for the school community.
P08	Head of Institution / Principal	Is developing confidence, initiative and a stronger sense of responsibility.
P09	Head of Institution / Principal	Participates in co-curricular activities with enthusiasm and commitment.
P10	Head of Institution / Principal	Continues to mature positively and should sustain these good habits.

1.14 Appendix C — How to use the PDF and workbook together

Use the PDF for execution order, roles, expected outcomes and acceptance decisions. Use the Excel workbook as a controlled reference/test dataset only. First compare its IDs and headings with the live NEXUS template; then copy a small approved subset into a QA-only workflow. The workbook intentionally labels supplemental fictional records because the read-only live audit does not contain the 480-learner configuration described in the operational guide. Never import the workbook wholesale into the existing school without approval.